MIL-E-1/736B
2 October 1968
SUPERSEDING
MIL-E-1/736A
18 June 1957

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MICROWAVE

TYPE 2B22

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Diode, UHF

See figure 1

Mounting position: Any

Weight: 1 ounce nominal

ABSOLUTE RATINGS:

Parameter:	Fl	Ef	Ib	ib (see note 1)	Eo	eb (see note 1)	epx	Ehk	tk	T(seal)
Unit:	MHz	Vac	mAdc	a	Vdc	v	v	Vdc	sec	° C
Maximum:	1,200	6.6	20	1.0	150	100	300	90	---	200
Minimum:	---	6.0	---	---	---	---	---	---	60	---
TEST CONDITIONS:	---	6.3	20	---	---	---	---	---	300	---

GENERAL:

Qualification - Required

Holding period (MIL-STD-1311) t = 168 hours

(B) denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:34Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
(B) ---	Visual and mechanical inspection criteria		---	---	---	---	---	---
3204	Voltage drop	Ib = 20 mAdc	0.65	II	Etd	3.0	9.0	Vdc
1231	Pulsing emission	is = 0.90 a; tp = 2 μs; prr = 500	0.65	II	eb	---	150	v
	Quality conformance inspection, part 2							
(B) ---	Dimensions	See figure 1	6.5	S3	---	---	---	---
1301	Heater current		6.5	S3	If	700	800	mAdc
(B) 1336	Heater-cathode leakage		6.5	S3	Ihk	---	50	μAdc
(B) 1331	Direct-interelectrode capacitance	Use standard socket in accordance with DESC Drawing 65025	6.5	S3	Cpk	1.9	2.4	pF
(B) 1211	Insulation of electrodes		6.5	S3	Rpsh	100	---	Meg
	Quality conformance inspection, part 3							
(B) ---	Life-test provisions	Group A; Rk/Ib = 12 mAdc; Epp = 30 Vac; t = 500 hours	---	---	---	---	---	---
---	Life-test end point	Voltage drop	---	---	Etd	---	10.0	Vdc
(B)	Periodic-check tests							
---	Dimensions	See figure 1	---	---	---	---	---	---
(B) 1101	Secureness of base, cap, or insert	Torque to be applied between anode and pins without shock	---	---	---	20	---	lb-in.

NOTE:

1. The tube shall not be operated more than 5 μs in a 100 μs interval.

Custodians:
Army - EL
Navy - EC
Air Force - 11

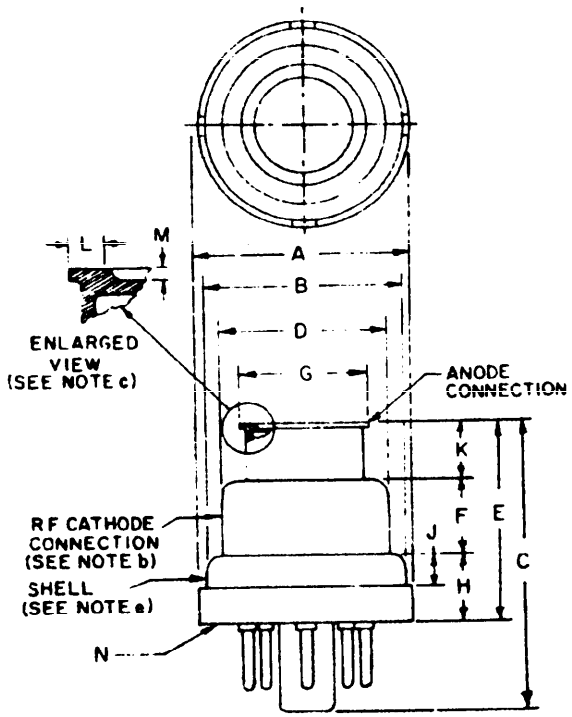
Review activities:
Army - EL
Navy -
Air Force - 11, 85
DSA - ES

User activities:
Army - MO
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

(Project 5960-2202-2)

Source: https://assist.dla.mil -- Downloaded: 202640-15T13:34Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note f)	
	Minimum	Maximum
Quality conformance inspection, part 2		
D	1.023 (25.98)	1.039 (26.39)
E		1.350 (34.29)
F	.475 (12.07)	.505 (12.83)
G	.806 (20.47)	.818 (20.78)
K	.360 (9.14)	.390 (9.91)
Quality conformance inspection, part 3 (periodic check)		
A		1.312 (33.32)
B	1.216 (30.89)	1.236 (31.39)
C		1.938 (49.23)
H	.395 (10.03)	.455 (11.56)
J	.180 (4.57)	.210 (5.33)
L	.063 (1.60)	
M	.013 (.33)	.027 (.69)
N	Base: B6-108	

Pin connections

Pin No.	Element
1	int con
2	h
3	k
5	k
7	h
8	k
shell	k (rf)
washer	a

NOTES:
a. Silver plate external surface of metal parts, except base pins, min .100 MSI.
b. Cathode RF connection and anode connection shall be concentric with respect to each other within .020.
c. Glass shall not extend beyond edge of anode.
d. Tolerance does not apply to point where shell is crimped.
e. Shell shall not be used for RF contact surface.
f. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Outline drawing of electron tube type 2B22.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:34Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:34Z
Check the source to verify that this is the current version before use.